GECG10069 (561085) F26: Introduction to Programming (C++)

Lab 7 : Functions

What you will learn from Lab 7

In this lab, you will learn different ways to pass and modify variables in functions, including call by value, using global variables, and call by reference.

TASK 7-1 : CALL BY VALUE V.S. GLOBAL VARIABLE V.S. CALL BY REFERENCE

✔ **Call By Value**: The values of the variables (x, y) in *main* are copied into the local variables (x, y) in the *swap* function.

➢ Any modification of the local variables (x, y) inside *swap* does **not** affect the original variables in *main*.

```
#include <iostream>
using namespace std;

void swap(int, int);
int main()
{
    int x = 5, y = 6;
    cout << "x = " << x << " and y = " << y << endl;
    swap(x, y);
    cout << "x = " << x << " and y = " << y << endl;
    return 0;
}
void swap(int x, int y)
{
    int temp;
    temp = x;
    x = y;
    y = temp;
}
```

✔ **Global Variable**: Variables are defined outside of all functions and can be accessed directly by any function.

➢ Modifications inside a function will **change the global variables permanently**, affecting all functions that use them.

```
#include <iostream>
using namespace std;

void swap();
int x = 5, y = 6;
int main()
{
```

```
    cout << "x = " << x << " and y = " << y << endl;
    swap();
    cout << "x = " << x << " and y = " << y << endl;
    return 0;
}
void swap()
{
    int temp;
    temp = x;
    x = y;
    y = temp;
}
```

✔ **Call By Reference**: The function receives the references (aliases) of the variables in *main*.

- Any modification inside the *swap* function **directly affects** the original variables in *main*.

```
#include <iostream>
using namespace std;

void swap(int&, int&);
int main()
{
    int x = 5, y = 6;
    cout << "x = " << x << " and y = " << y << endl;
    swap(x, y);
    cout << "x = " << x << " and y = " << y << endl;
    return 0;
}
void swap(int& x, int & y)
{
    int temp;
    temp = x;
    x = y;
    y = temp;
}
```

Exercise 7-1 : Texas Poker

Description -

WeiWei and MengMeng love playing Texas Poker. They want to write a program that simulates the four betting rounds of a Texas Poker.
In each round the dealer deals the cards for that round, then you decide what to do.

The rounds and the cards dealt are:

round	name	cards dealt
0	Pre-flop	2
1	Flop	3
2	Turn	1
3	River	1

Each card is a random suit (spade, heart, diamond, club) and a random number from 1 to 13.

Input :

- You are given two files named **input1.txt** and **input2.txt** containing up to four commands separated by spaces, one for each round. Each command is raise, call, or fold.

Output :

For each round, print the cards dealt, one card per line, then print the pot.
The pot starts at 10.

- raise — the pot is doubled
- call — the pot does not change
- fold — print you fold, the pot becomes 0, and the hand ends immediately

Constraints :

- *You must finish the functions:*

```
int card_num(); //output a random number between 1 to 13
string card_flower(); // output a random suit (spade, heart, diamond, club)
void dealer(int round);  //deal card based on different round
int pot_num(string in, int pot_now);  // cout the total pot
```

- *You may ignore the repeated cards.*
- *Random seed fix at 0. Use srand(0).*

Sample Code-

```
…

int pot_num(string in, int pot_now) {
   int pot_new = pot_now;

   //update pot
   ……

   return pot_new;
}

int card_num() {
   //output a random number between 1 to 13
   ……
}

string card_flower() {
  //Randomly output string: "spade", "heart", "diamond", or "club"
   ……
}

void dealer(int round) {
   if (round == 0) {
      //deal two cards
   }
   else if (round == 1) {
      //deal three cards
   }
   else if (round == 2) {
      //deal one card
```

```
    }
    else if (round == 3) {
        //deal one card
    }
}

int main() {
    srand(0);

    int pot = 10;
    string in;

    for (int round = 0; round < 4; round++) {
        //deal cards
        ……

        cin >> in;
        //count pots
        ……
    }

    ……

    return 0;
}
```

Sample Testcases -

Sample Input - 1
call call call call
Sample Output - 1
heart 5 **spade 4** **pot: 10** **heart 9**

heart 3
club 13
pot: 10
spade 11
pot: 10
spade 7
pot: 10
total pot: 10

Sample Input - 2
raise fold
Sample Output - 2
diamond 11 diamond 1 pot: 20 club 7 diamond 1 club 1 you fold total pot: 0

Exercise 7-2 : Counting Poker Hands

Description -

A standard deck has 52 cards, in Texas poker players combine two private hole cards with five shared community cards to make the best five-card hand.

The problem is how many different 5-card hands there are?

More generally, write a program that counts how many ways there are to choose k cards out of n.

Input -

- You are given a file named input3.txt containing 5 lines of two integers n and k, separated by a space.

Output -

- Print a single integer — the number of different ways to choose k cards out of n.

Hint -

Think about the first card in the deck. There are only two possibilities:

- you take it, and then you still need k - 1 cards out of the remaining n - 1
- you leave it, and then you still need k cards out of the remaining n - 1
- Consider Pascal's Identity

Constrains -

- 1 <= n <= 52
- 0 <= k <= 5
- k <= n
- You must finish the function:

```
int comb(int n, int k);
```

Sample Testcases -

Sample Input - 1
52 5
Sample Output - 1
2598960

Sample Input - 2
13 2
Sample Output - 2

78

Sample Input - 3
5 5
Sample Output - 3
1